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Business Architecture Meta Model (BAMM) Request For Proposal

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Letters of Intent due: 7 December, 2015
Submissions due: May 15, 2016

Objective of this RFP

There are a number of different graphical and tabular models in use by business architects for capture, development and communication of top management strategy, analysis, planning and business requirements. These models are typically implemented with basic office applications such as spreadsheets, and graphical drawing tools. A set of complementary models used together will share some of the same concepts, but they are implemented independently so they may be inconsistent and development and evolution of the models is very tedious. They are missing computer assistance in the consistent development of an integrated model.

This RFP calls for a business architecture metamodel that defines the concepts and relationships that support generally accepted business architect and management planning models as views of an integrated mode. This metamodel shall support the development of a business architecture model based on BMM and it shall be compatible with VDML, BPMN, CMMN and DMN since models in these languages will be used to develop the more detailed business models.

It is not the purpose of this RFP to define a standard set of management and business architect views, but rather to identify the concepts and relationships that collectively occur in existing views so that implementations can support and develop preferred sets of views for use by business leaders and business architects.

For further details see Section 6 of this document.

6 Specific Requirements on Proposals

This section defines requirements for proposals that are specific to this RFP.

6.1 Problem Statement

Business architects provide the link between business strategy and business design based on business objectives, best practices, capabilities, relevant technologies, constraints and stakeholder values. They promote the development of consistent and appropriate business design for a shared understanding of the business strategy and transformation requirements. The number of different models required, the complexity of analysis and the need to provide a clear specification of a desired future design of the enterprise all require an integrated, computer-based model. A business architecture metamodel that comprehends the concepts and relationships depicted in current management and business architect tabular and graphical models will provide the structure for a shared computer model that ensures the consistency of multiple business architecture models.

This integrated model shall be compatible with BMM to support development of an integrated model based on a BMM strategic plan. It shall be compatible with VDML, BPMN, CMMN and DMN to enable these languages to be used to develop more robust models of the to-be business design.

6.2 Scope of Proposals Sought

This RFP solicits proposals for a metamodel that defines the elements of a variety of business architecture models in a business management viewpoint. This metamodel will provide a level of abstraction suitable for business architects to capture and express the patterns and requirements of the current or future design of the enterprise. As such, it shall provide a level of abstraction most appropriate for providing more detail for strategic plans and models of top-level management. These models shall be aligned with existing OMG standards such as BMM, while providing input to more detailed modelling, potentially using VDML, BPMN, CMMN and/or DMN.

A number of existing modelling techniques and forms of presentation are useful for business architecture modelling and for presentation and discussion of solutions with business leaders. Currently, many of these models are created independently and may not represent a consistent solution. It is not the purpose of this RFP to define a standard set of views, but rather to develop a meta model that identifies the concepts and relationships that collectively occur in existing views so that implementations can support and develop preferred sets of views for use by business architects.

A submission shall demonstrate that the metamodel supports a representative set of management and business architect models in order to establish that is it is sufficient to support current and potential, future management and business architect views. A submitter shall identify the views of the representative set and show a mapping of the model meta object elements to the integrated metamodel. The specification is not expected to include the graphical and/or tabular displays, but such specifications are optional. It is expected that the existing views are representative of the information of interest, but when an integrated model is available, new and better views will be developed. A normative requirement to support an existing set of display formats would stifle market incentives to innovate.

6.3 Relationship to other OMG Specifications and activities

6.3.1 Relationship to OMG specifications

UML. The notation for expression of the BAMM metamodel will be the MOF subset of UML. http://www.omg.org/spec/UML

MOF. The BAMM specification metamodel must be expressed as a MOF metamodels. http://www.omg.org/spec/MOF

XMI. Import and export of BAMM models shall comply with XMI. http://www.omg.org/spec/XMI

BMM. The BAMM specification shall be compatible with BMM. http://www.omg.org/spec/BMM

VDML. The BAMM specification shall be compatible with VDML. http://www.omg.org/spec/VDML/

SMM. The BAMM specification shall incorporate SMM for representation of measurements and mathematical computations VDML. http://www.omg.org/spec/SMM

BPMN. The BAMM specification shall be compatible with BPMN. http://www.omg.org/spec/BPMN/index.htm

CMMN. The BAMM specification shall be compatible with CMMN. http://www.omg.org/spec/CMMN

6.3.2 Relationship to other OMG Documents and work in progress

The OMG P&P requires consistency with existing OMG specifications.

6.4 Related non-OMG Activities, Documents and Standards

The Business Architecture Guild has work on a business architecture metamodel in process.

The Global University Alliance has done extensive research on current business practices and has identified management views, concepts and relationships.

6.5 Mandatory Requirements

6.5.1 MOF metamodel

Submissions shall specify the business architecture metamodel (BAMM) as a MOF metamodel.

6.5.2 Scope of meta objects involved

Submissions shall use the list of business architecture meta object concepts described in Appendix A.2 of this RFP as a basis for determining the scope of this metamodel.

6.5.3 Support for business architecture artefacts

Submissions shall show support a comprehensive set of graphical and tabular business architecture artefacts currently in use for development, capture and communication of business architecture, but the graphical representations shall not be required for compliance.

6.5.4 Alignment with VDML, BPMN, CMMN and DMN

Submissions shall show semantic alignment with appropriate VDML, BPMN, CMMN and DMN concepts.

6.5.5 Alignment with BMM

BAMM shall show semantic alignment with appropriate BMM concepts as a basis for modelling strategic planning initiatives.

6.6 Non-mandatory features

6.6.1 Notation

Graphical and/or tabular notation shall be acceptable to ensure consistency, but shall be optional for compliance in order to enable advances in notation that make better use of the integrated model.

6.7 Issues to be discussed

6.7.1 Mapping of business architecture artefacts and models to BAMM concepts

Submitters shall identify the elements of representative management and business architecture models and demonstrate how these are supported by the submission.

6.8 Relationship to BMM

Submitters shall describe how users will apply BMM with BAMM.

6.9 Relationships VDML, BPMN, CMMN and DMN

Submitters shall describe how users will apply BAMM with VDML, BPMN CMMN and/or DMN.

6.10 Evaluation Criteria

Semantic alignment with related OMG modelling languages—BMM, VDML, BPMN, CMMN and DMN.

Completeness of concepts and relationships to support existing tabular and graphical business architecture models for business design.

6.11 Other information unique to this RFP

The OMG BA SIG, has together with the OMG Academia & Research Working Group, initiated a world wide Business Architecture research and analysis effort together with the Global University Alliance (GUA)—which consist of 450 universities, professors, researchers and lecturers, and the standards organizations ISO, IEEE, CSIR, and LEADing Practice. The findings of the research and analysis are the basis of some of the requirements found in this RFP. For more information see: http://www.globaluniversityalliance.net/research-areas/business-architecture-research/ IPR Mode

6.12 IPR Mode

Every OMG Member that makes any written Submission in response to this RFP shall provide the Non-Assertion Covenant found in Appendix A of the OMG IPR Policy [IPR].

6.13 RFP Timetable

The timetable for this RFP is given below. Note that the TF or its parent TC may, in certain circumstances, extend deadlines while the RFP is running, or may elect to have more than one Revised Submission step. The latest timetable can always be found at the OMG *Work In Progress* page at http://www.omg.org/schedules under the item identified by the name of this RFP.

Event or Activity	Date
Letter of Intent (LOI) deadline	*7 December 2016*
Initial Submission deadline	*15 May 2016*
Voter registration closes	*1 November 2016*
Initial Submission presentations	*x June 2016*
Revised Submission deadline	*15 November 2016*
Revised Submission presentations	*x December 2016*

Appendix A References & Glossary Specific to this RFP

A.1 References Specific to this RFP

http://www.globaluniversityalliance.net/research-areas/business-architecture-research/

A.2 Glossary Specific to this RFP

The terms defined below identify a set of concepts that are associated with models used by business leaders and business architects in the development and communication of strategic plans and business architecture. They are the result of extensive research into current business practices by the Global University Alliance. They are included here to illustrate the expected scope of the Business Architecture Meta Model.

Meta Object	Description
Actor	Any person, organization, or system that many be assigned one or more roles. Actors may be internal or external to an organization.
Application Role	A participation recognized within an application, providing behaviour automate or enable some parts of a business function, service or process task.
Application Service	An externally visible unit of functionality, provided by one or more components, exposed through well-defined interfaces, and meaningf to the environment.
Application Task	The automated behaviour of a process activity performed by an application.
Business Object	A enterprise world thing which relates to the enterprise's means to a
Business Rule	A statement that defines or constrains some aspect of behaviour wit the enterprise and always resolves to either true or false.
Business Workflow	A flow, stream, sequence, course, succession, series or progression as well as order for the movement of information or material from on enterprise function, service or activity (work site) to another.
Contract	An agreement between two or more parties that establishes conditio for interaction.
Control	The exercise of restraining or directing influence. It includes decisio making aspects with accompanying decision logic necessary to ensu compliance.
Data Object	A logical cluster of related data representing the data object view of business or information object.
Data Service	A standardized and uniform way of accessing information in a form t is useful to enterprise applications without requiring knowledge of its physical persistence structure.
Driver	An external or internal factor that drives, establishes motivation for o influences the direction of an enterprise.
Enterprise Capability	An enterprise capability is an abstraction that represents the ability t perform a particular type of work.

Event	Something that happens, this may include a planned occasion or a state change that effects the triggering or termination of processing.
Force (external/ internal)	An external or internal factor that forces or pushes some aspect of a enterprise in a specific direction.
Gateway	Determines forking and merging of paths depending on the conditior expressed.
Goal	A desired aspiration considered a part of the organizational directior
Information Object	Information about enterprise world objects that can be in any mediur form.
Location	A point, facility, place or geographic position that may be referred to physically or logically.
Measure	Any type of metric used to gauge some quantifiable component of ar enterprise's performance.
Mission	The purpose and nature of the enterprise.
Objective	The purpose or target of one's efforts or actions.
Organizational Function	A cluster of tasks performing a specific class of jobs.
Plan	A ordered specification of activities, resources and deliverables to achieve a desired result.
Planning	The notion of thinking about and organizing the tasks required to achieve a desired output.
Process	A set of structured activities or tasks with logical behaviour that produces a specific service or product.
Product	A result and output generated by the enterprise. It has a combinatior tangible and intangible attributes (i.e.. features, functions, usage).
Quality	A state of excellence or worth specifying the essential and distinguishing individual nature and the attributes based on the inten use.
Report	The exposure, description, and portrayal of information about the status, direction or execution of work within the functions, services, processes, and resources of the enterprise.

Resource	A specific person, expertise, data, information, material, machine, la capital or organization that is required to accomplish an activity or as means to act on behalf of the enterprise to achieve a desired outcom
Risk	The combined impact of any conditions or events, including those caused by uncertainty, change, hazards or other factors that can adversely affect the potential for achieving objectives.
Role	A participation that something or someone has the rights, rules, competencies, and capabilities to perform. A resource and/ or actor may have a number of roles i.e. process role, service role or applicat role and many actors may be assigned the same role.
Security	The objects or tools that secure, make safe and protect through measures to prevent exposure to danger or risk.
Service (Business Service)	The externally visible [logical] deed or effort performed to satisfy a ne or to fulfill a demand that is meaningful to the [business] environmen
Strategy	The direction and ends to which the enterprise seeks as well as the means and methods by which the ends will be attained.
Time	A point in time, duration, period in time, or time sequence for when something should start, take place or be completed or the amount of time consumed.
Value Proposition	The merit and benefit that a customer, added value partner or the market itself is expected to obtain from their perspective and point of view.
Vision	The desired future state of the enterprise. An imagination of the futu aspirational state of how the enterprise could or should be viewed without regard as to how this will be achieved.

Appendix B General Reference and Glossary

B.1 General References

The following documents are referenced in this document:

[BCQ] OMG Board of Directors Business Committee Questionnaire, *http://doc.omg.org/bcq*

[CCM] CORBA Core Components Specification
http://www.omg.org/spec/CCM/

[CORBA] Common Object Request Broker Architecture (CORBA)
http://www.omg.org/spec/CORBA/

[CORP] UML Profile for CORBA,
http://www.omg.org/spec/CORP

[CWM] Common Warehouse Metamodel Specification
http://www.omg.org/spec/CWM

[EDOC] UML Profile for EDOC Specification
http://www.omg.org/spec/EDOC/

[Guide] The OMG Hitchhiker's Guide
http://doc.omg.org/hh

[IDL] Interface Definition Language Specification
http://www.omg.org/spec/IDL35

[INVENT] Inventory of Files for a Submission/Revision/Finalization
http://doc.omg.org/inventory

[IPR] IPR Policy
http://doc.omg.org/ipr

[ISO2] ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards
http://isotc.iso.org/livelink/livelink?func=ll&objId=4230456

[LOI] OMG RFP Letter of Intent template
http://doc.omg.org/loi

[MDAa] OMG Architecture Board, "Model Driven Architecture - A Technical Perspective"
http://www.omg.org/mda/papers.htm

[MDAb] Developing in OMG's Model Driven Architecture (MDA)
http://www.omg.org/mda/papers.htm

[MDAc] MDA Guide
http://www.omg.org/docs/omg/03-06-01.pdf

[MDAd] MDA "The Architecture of Choice for a Changing World
http://www.omg.org/mda

[MOF] Meta Object Facility Specification
http://www.omg.org/spec/MOF/

[NS] Naming Service
http://www.omg.org/spec/NAM

[OMA] Object Management Architecture
http://www.omg.org/oma/

[OTS] Transaction Service
http://www.omg.org/spec/OTS

[P&P] Policies and Procedures of the OMG Technical Process
http://doc.omg.org/pp

[RAD] Resource Access Decision Facility
http://www.omg.org/spec/RAD

[ISO2] ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards
http://isotc.iso.org/livelink/livelink?func=ll&objId=4230456

[RM-ODP]
ISO/IEC 10746

[SEC] CORBA Security Service
http://www.omg.org/spec/SEC

[TEMPL] Specification Template
http://doc.omg.org/submission-template

[TOS] Trading Object Service
hptp://www.omg.org/spec/TRADE

[UML] Unified Modeling Language Specification,
http://www.omg.org/spec/UML

[XMI] XML Metadata Interchange Specification,
http://www.omg.org/spec/XMI

B.2 General Glossary

Architecture Board (AB) - The OMG plenary that is responsible for ensuring the technical merit and MDA-compliance of RFPs and their submissions.

Board of Directors (BoD) - The OMG body that is responsible for adopting technology.

Common Object Request Broker Architecture (CORBA) - An OMG distributed computing platform specification that is independent of implementation languages.

Common Warehouse Metamodel (CWM) - An OMG specification for data repository integration.

CORBA Component Model (CCM) - An OMG specification for an implementation language independent distributed component model.

Interface Definition Language (IDL) - An OMG and ISO standard language for specifying interfaces and associated data structures.

Letter of Intent (LOI) - A letter submitted to the OMG BoD's Business Committee signed by an officer of an organization signifying its intent to respond to the RFP and confirming the organization's willingness to comply with OMG's terms and conditions, and commercial availability requirements.

Mapping - Specification of a mechanism for transforming the elements of a model conforming to a particular metamodel into elements of another model that conforms to another (possibly the same) metamodel.

Metadata - Data that represents models. For example, a UML model; a CORBA object model expressed in IDL; and a relational database schema expressed using CWM.

Metamodel - A model of models.

Meta Object Facility (MOF) - An OMG standard, closely related to UML, that enables metadata management and language definition.

Model - A formal specification of the function, structure and/or behavior of an application or system.

Model Driven Architecture (MDA) - An approach to IT system specification that separates the specification of functionality from the specification of the implementation of that functionality on a specific technology platform.

Normative – Provisions to which an implementation shall conform to in order to claim compliance with the standard (as opposed to non-normative or informative material, included only to assist in understanding the standard).

Normative Reference – References to documents that contain provisions to which an implementation shall conform to in order to claim compliance with the standard.

Platform - A set of subsystems/technologies that provide a coherent set of functionality through interfaces and specified usage patterns that any subsystem that depends on the platform can use without concern for the details of how the functionality provided by the platform is implemented.

Platform Independent Model (PIM) - A model of a subsystem that contains no information specific to the platform, or the technology that is used to realize it.

Platform Specific Model (PSM) - A model of a subsystem that includes information about the specific technology that is used in the realization of it on a specific platform, and hence possibly contains elements that are specific to the platform.

Request for Information (RFI) - A general request to industry, academia, and any other interested parties to submit information about a particular technology area to one of the OMG's Technology Committee subgroups.

Request for Proposal (RFP) - A document requesting OMG members to submit proposals to an OMG Technology Committee.

Task Force (TF) - The OMG Technology Committee subgroup responsible for issuing a RFP and evaluating submission(s).

Technology Committee (TC) - The body responsible for recommending technologies for adoption to the BoD. There are two TCs in OMG – the *Platform TC* (PTC) focuses on IT and modeling infrastructure related standards; while the *Domain TC* (DTC) focuses on domain specific standards.

Unified Modeling Language (UML) - An OMG standard language for specifying the structure and behavior of systems. The standard defines an abstract syntax and a graphical concrete syntax.

UML Profile - A standardized set of extensions and constraints that tailors UML to particular use.

XML Metadata Interchange (XMI) - An OMG standard that facilitates interchange of models via XML documents.

<Note to RFP Editors: Append additional appendices if needed here and update the list and brief description of appendices in Section 1.>